

Analysis 7: Causal Confounder Adjustment

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0.1 Overview

This report estimates the confounder-adjusted effect of early antibiotic administration (within 3 hours of ICU admission, per the Surviving Sepsis Campaign hour-3 bundle threshold) on in-hospital mortality among Sepsis-3 patients. Sicker patients may receive antibiotics faster (or slower, if sepsis is not yet recognised), so a naive comparison of early- versus delayed-antibiotic mortality is biased by indication. Four estimators of increasing robustness are compared: (1) crude unadjusted, (2) propensity-score matching, (3) inverse-probability weighting, and (4) augmented-IPW (doubly robust). The adjustment set is chosen by LASSO rather than by hand.

0.2 Load Libraries

```
suppressPackageStartupMessages({  
  library(glmnet); library(MatchIt); library(WeightIt); library(cobalt); library(sandwich)  
})  
set.seed(486649)
```

0.3 Locate and Read Data

```
find_data <- function(f) {  
  cands <- c(file.path("../..//processed_data", f),  
             file.path("processed_data", f), f)  
}
```

```

hit <- cands[file.exists(cands)][1]
if (is.na(hit)) stop("cannot find ", f); hit
}
OUT <- dirname(find_data("05_analysis_matrix.parquet"))
FIG <- if (dir.exists("figure")) "figure" else { dir.create("figure"); "figure" }

am <- as.data.frame(arrow::read_parquet(find_data("05_analysis_matrix.parquet")))
si <- as.data.frame(arrow::read_parquet(find_data("03_suspected_infection.parquet")))
# Explicit format prevents silent date-only fallback when any element is malformed.
parse_dt <- function(x) as.POSIXct(x, format = "%Y-%m-%d %H:%M:%S", tz = "UTC")
am$intime <- parse_dt(am$intime)
si$t_abx <- parse_dt(si$t_abx)

```

0.4 Build the Sepsis-3 Cohort with Antibiotic Timing

```

d <- am[am$sepsis3 == "True" | am$sepsis3 == TRUE, ]
d <- merge(d, si[, c("stay_id", "t_abx")], by = "stay_id")
d$hours_to_abx <- as.numeric(difftime(d$t_abx, d$intime, units = "hours"))
d$early_abx <- as.integer(d$hours_to_abx <= 3)
d$mortality <- d$hospital_expire_flag
cat(sprintf("Sepsis-3 cohort with antibiotic timing: %d stays (%.0f%% early <=3h), mortality %.1f%%\n",
            nrow(d), 100 * mean(d$early_abx), 100 * mean(d$mortality)))

```

Sepsis-3 cohort with antibiotic timing: 24203 stays (48% early <=3h), mortality 18.7%

0.5 Collapse High-Cardinality Covariates

Admission type, care unit, and race are collapsed into clinically sensible groups to avoid near-empty dummy categories in matching and weighting.

```

d$admission_grp <- with(d, ifelse(grepl("EMER|URGENT", admission_type), "EMERGENT",
                                   ifelse(grepl("ELECTIVE|SURGICAL SAME DAY", admission_type), "ELECTIVE",
                                           "OBSERVATION")))
d$careunit_grp <- with(d, ifelse(grepl("Medical", first_careunit), "MEDICAL",
                                   ifelse(grepl("Cardiac|Coronary", first_careunit), "CARDIAC",
                                           ifelse(grepl("Neuro", first_careunit), "NEURO",
                                                  ifelse(grepl("Surg|Trauma", first_careunit), "SURGICAL", "OTHER")))))
top_race <- names(sort(table(d$race), decreasing = TRUE))[1:4]
d$race_grp <- ifelse(d$race %in% top_race, d$race, "OTHER/UNKNOWN")
for (v in c("admission_grp", "careunit_grp", "race_grp"))
  d[[v]] <- factor(d[[v]])
d$female <- as.integer(d$gender == "F")

```

0.6 LASSO Covariate Selection

Following Guo et al., LASSO penalised logistic regression on the outcome model selects confounders from a candidate set. The exposure `early_abx` is forced in unpenalised (`penalty.factor=0`).

```

cand_form <- ~ early_abx + age + female + admission_grp + careunit_grp + race_grp +
  sofa_resp + sofa_coag + sofa_liver + sofa_cardio + sofa_cns + sofa_renal
Xcand <- model.matrix(cand_form, data = d)[, -1] # drop intercept
pen <- ifelse(colnames(Xcand) == "early_abx", 0, 1)
cv <- cv.glmnet(Xcand, d$mortality, family = "binomial", alpha = 1,
  penalty.factor = pen, nfolds = 10)
sel <- rownames(coef(cv, s = "lambda.min"))[which(coef(cv, s = "lambda.min")[, 1] != 0)]
sel <- setdiff(sel, c("(Intercept)", "early_abx"))
# Map dummy columns back to parent covariates for matching/weighting formulas.
parents <- c("age", "female", "admission_grp", "careunit_grp", "race_grp",
  "sofa_resp", "sofa_coag", "sofa_liver", "sofa_cardio", "sofa_cns", "sofa_renal")
kept <- parents[sapply(parents, function(p) any(grepl(paste0("^", p), sel)) || p %in% sel)]
if (length(kept) < 3) kept <- parents # fallback: LASSO pruned too aggressively
cat(sprintf("\nLASSO-selected confounder set (%d/%d candidates): %s\n",
  length(kept), length(parents), paste(kept, collapse = ", ")))

##
## LASSO-selected confounder set (11/11 candidates): age, female, admission_grp, careunit_grp, race_grp, sofa_resp, sofa_coag, sofa_liver, sofa_cardio, sofa_cns, sofa_renal
adj_form <- as.formula(paste("early_abx ~", paste(kept, collapse = " + ")))

```

0.7 Crude (Unadjusted) Estimate

```

crude_glm <- glm(mortality ~ early_abx, data = d, family = binomial)
crude_or <- exp(coef(crude_glm)["early_abx"])
crude_rd <- mean(d$mortality[d$early_abx == 1]) - mean(d$mortality[d$early_abx == 0])

```

0.8 Propensity-Score Matching

1:1 nearest-neighbour matching with a 0.2 SD caliper. The ATT is estimated with cluster-robust standard errors grouped by matched pair.

```

m.out <- matchit(adj_form, data = d, method = "nearest", distance = "glm", caliper = 0.2)
md <- match.data(m.out)
psm_glm <- glm(mortality ~ early_abx, data = md, weights = weights, family = binomial)
psm_se <- sqrt(diag(vcovCL(psm_glm, cluster = md$subclass)))
psm_or <- exp(coef(psm_glm)["early_abx"])
psm_ci <- exp(coef(psm_glm)["early_abx"] + c(-1.96, 1.96) * psm_se["early_abx"])

```

0.9 Covariate Balance Diagnostics

```

bal <- bal.tab(m.out, un = TRUE, m.threshold = 0.1)
bal_df <- as.data.frame(bal$Balance)
bal_df$covariate <- rownames(bal_df)
write.csv(bal_df[, c("covariate", grep("Diff", names(bal_df), value = TRUE))],
  file.path(OUT, "19_analysis7_covariate_balance.csv"), row.names = FALSE)
png(file.path(FIG, "19_analysis7_balance_love_plot.png"), width = 1200, height = 900, res = 150)
print(love.plot(m.out, stats = "mean.diffs", threshold = 0.1, abs = TRUE,
  var.order = "unadjusted",

```

```

        title = "Covariate balance: before vs after PS matching"))
dev.off()

```

```

## pdf
## 2

```

0.10 Inverse-Probability Weighting

```

w.out <- weightit(adj_form, data = d, method = "ps", estimand = "ATE")
ipw_glm <- glm(mortality ~ early_abx, data = d, weights = w.out$weights, family = binomial)
ipw_se <- sqrt(diag(vcovHC(ipw_glm, type = "HCO")))
ipw_or <- exp(coef(ipw_glm)["early_abx"])
ipw_ci <- exp(coef(ipw_glm)["early_abx"] + c(-1.96, 1.96) * ipw_se["early_abx"])

```

0.11 Doubly Robust / Augmented-IPW Estimator

The AIPW estimator is consistent for the ATE if either the propensity model or the outcome model is correctly specified (van der Laan and Robins, 2003). Separate outcome regressions under treatment and control are combined with propensity weights. A nonparametric bootstrap (200 replicates) provides the confidence interval, refitting all three models in each replicate.

```

ps <- w.out$ps
outcome_form <- as.formula(paste("mortality ~", paste(kept, collapse = " + ")))
mu1_fit <- glm(outcome_form, data = d[d$early_abx == 1, ], family = binomial)
mu0_fit <- glm(outcome_form, data = d[d$early_abx == 0, ], family = binomial)
mu1 <- predict(mu1_fit, newdata = d, type = "response")
mu0 <- predict(mu0_fit, newdata = d, type = "response")
aipw_est <- function(dd, ps_, mu1_, mu0_) {
  psi1 <- mean(mu1_ + dd$early_abx * (dd$mortality - mu1_) / ps_)
  psi0 <- mean(mu0_ + (1 - dd$early_abx) * (dd$mortality - mu0_) / (1 - ps_))
  psi1 - psi0
}
aipw_rd <- aipw_est(d, ps, mu1, mu0)

cat("bootstrapping AIPW CI (200 reps) ...\n")

```

```

## bootstrapping AIPW CI (200 reps) ...

```

```

B <- 200
boot_rd <- numeric(B)
n <- nrow(d)
for (b in 1:B) {
  idx <- sample(n, n, replace = TRUE)
  db <- d[idx, ]
  wb <- tryCatch(weightit(adj_form, data = db, method = "ps", estimand = "ATE"),
    error = function(e) NULL)
  if (is.null(wb)) { boot_rd[b] <- NA; next }
  m1b <- tryCatch(glm(outcome_form, data = db[db$early_abx == 1, ], family = binomial),
    error = function(e) NULL)
  m0b <- tryCatch(glm(outcome_form, data = db[db$early_abx == 0, ], family = binomial),

```

```

        error = function(e) NULL)
  if (is.null(m1b) || is.null(m0b)) { boot_rd[b] <- NA; next }
  boot_rd[b] <- aipw_est(db, wb$ps, predict(m1b, newdata = db, type = "response"),
                        predict(m0b, newdata = db, type = "response"))
}
aipw_ci <- quantile(boot_rd, c(0.025, 0.975), na.rm = TRUE)

```

0.12 Summary of Effect Estimates

```

effects <- data.frame(
  method = c("Crude (unadjusted)", "PSM (1:1 nearest, ATT)",
             "IPW (ATE weights)", "AIPW / doubly robust (ATE)"),
  estimand = c("RD & OR", "OR (matched)", "OR (weighted)", "RD"),
  risk_difference = c(crude_rd, NA, NA, aipw_rd),
  rd_ci_low = c(NA, NA, NA, aipw_ci[1]), rd_ci_high = c(NA, NA, NA, aipw_ci[2]),
  odds_ratio = c(crude_or, psm_or, ipw_or, NA),
  or_ci_low = c(NA, psm_ci[1], ipw_ci[1], NA),
  or_ci_high = c(NA, psm_ci[2], ipw_ci[2], NA))
effects[, -(1:2)] <- round(effects[, -(1:2)], 4)
cat("\nAntibiotic-timing effect on in-hospital mortality, by method:\n")

##
## Antibiotic-timing effect on in-hospital mortality, by method:

print(effects, row.names = FALSE)

##           method      estimand risk_difference rd_ci_low rd_ci_high
##      Crude (unadjusted)      RD & OR      -0.0187         NA         NA
##      PSM (1:1 nearest, ATT)  OR (matched)         NA         NA         NA
##      IPW (ATE weights) OR (weighted)         NA         NA         NA
##      AIPW / doubly robust (ATE)      RD      -0.0191    -0.0284    -0.0107
##      odds_ratio or_ci_low or_ci_high
##      0.8840         NA         NA
##      0.8994      0.8410      0.9617
##      0.8894      0.8325      0.9501
##      NA         NA         NA

write.csv(effects, file.path(OUT, "19_analysis7_effect_estimates.csv"), row.names = FALSE)

cat("\nAnalysis 7 (confounder-adjusted causal analysis) complete.\n")

##
## Analysis 7 (confounder-adjusted causal analysis) complete.

```